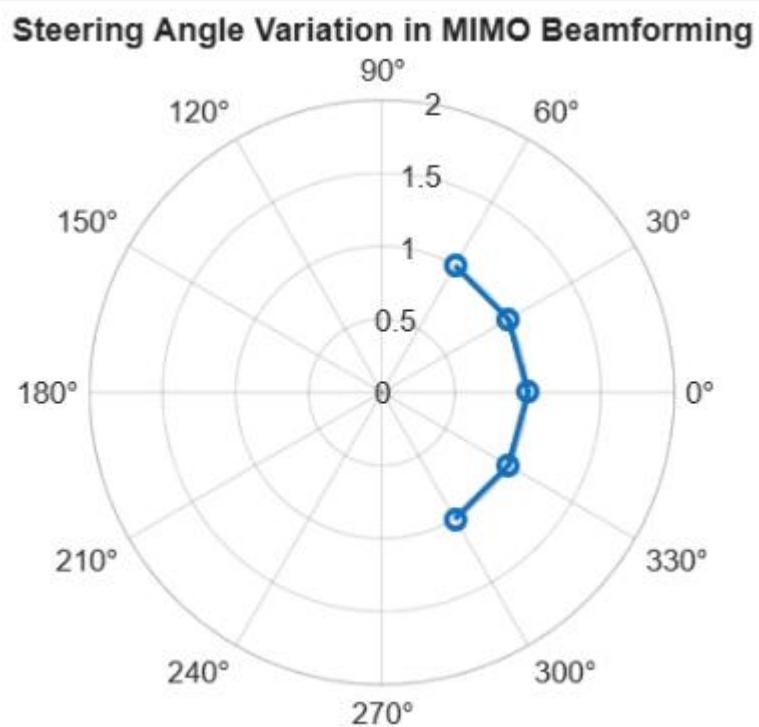


Experiment: 8

Steering Angle Variation in MIMO Beamforming Using MATLAB

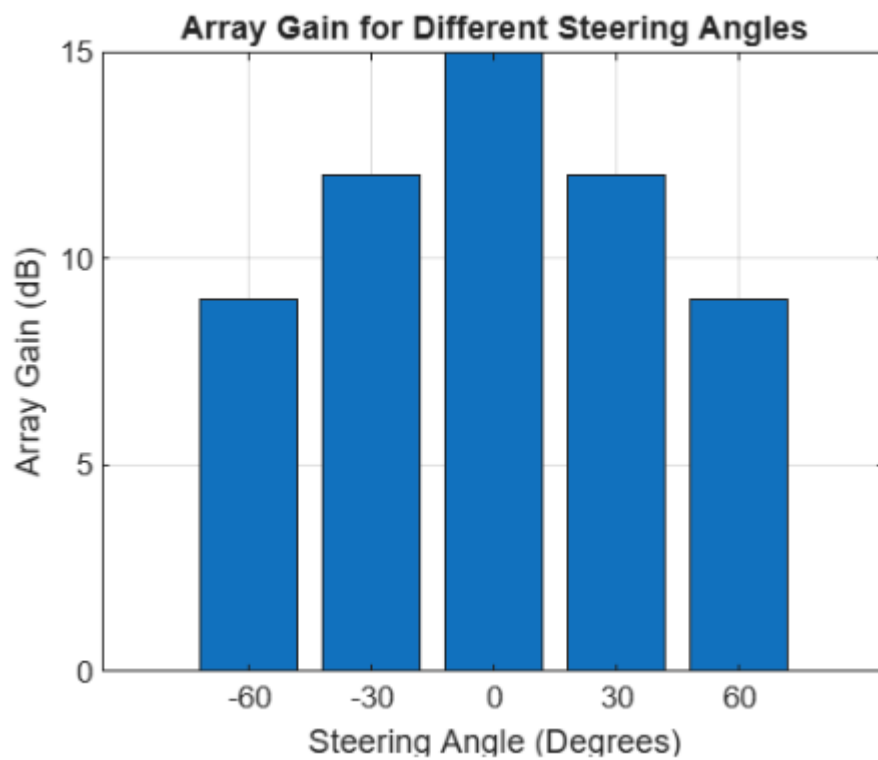
Objective – 1

```
clc;  
clear;  
close all;  
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)  
gain = ones(size(angle)); % Normalized Gain  
figure  
polarplot(deg2rad(angle), gain, 'o-', 'LineWidth', 2)  
title('Steering Angle Variation in MIMO Beamforming')
```



Objective – 2

```
clc;  
clear;  
close all;  
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)  
gain = [9 12 15 12 9]; % Array Gain (dB)  
figure  
bar(angle, gain)  
grid on  
xlabel('Steering Angle (Degrees)')  
ylabel('Array Gain (dB)')  
title('Array Gain for Different Steering Angles')
```



Objective – 3

```
clc;

clear;

close all;

% Number of Antenna Elements

antennas = [4 8 16 32 64];

% Antenna Gain (dB)

gain = 10 * log10(antennas);

% Plot Bar Graph

figure;

bar(antennas, gain);

grid on;

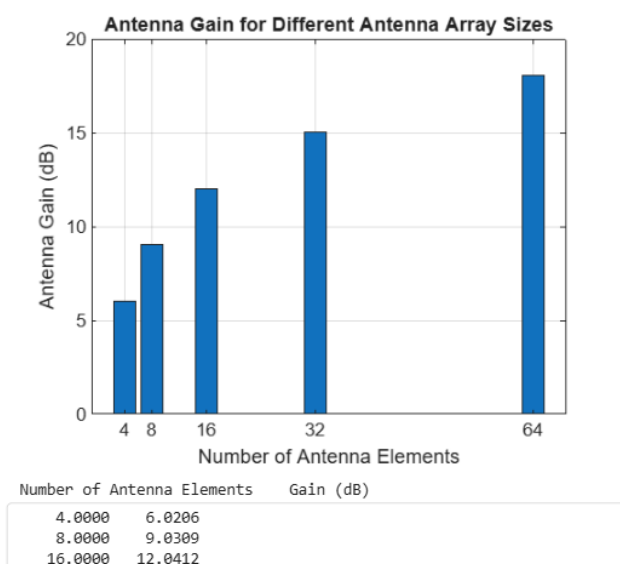
xlabel('Number of Antenna Elements');

ylabel('Antenna Gain (dB)');

title('Antenna Gain for Different Antenna Array Sizes');

% Display values in Command Window

disp('Number of Antenna Elements   Gain (dB)');disp([antennas' gain'
```



```
]);
```